When Does Temporal Abstraction Break World Models? An Empirical Investigation of DreamerV3

Anonymous Authors

Abstract

Temporal abstraction—imagining at variable time scales—is a natural extension for world-model reinforcement learning, yet practical implementations consistently underperform fixed-rate baselines. We present a controlled empirical investigation asking *when and why* variable-rate imagination fails in DreamerV3. Through a factorial ablation framework across four DeepMind Control environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop) with 3 seeds per condition, we find that the failure mode is *not* universal: pathological value overestimation (normalized value/eval ratio $>1.5\times$ baseline) and catastrophic collapse occur only in Walker Walk (2/3 seeds exceeding the threshold, 1/3 collapsing), while Cheetah Run, Cartpole Swingup, and Hopper Hop show uniform performance degradation without pathological overestimation (0/9 seeds exceeding $1.5\times$). We trace this divergence to environment-specific RSSM prediction fidelity: Walker's observation prediction gap at Δt=2 is $2.04\times$ (vs. $1.35\times$ for Cheetah), creating sufficient model error for trained Δt-embeddings to amplify into a delusional critic feedback loop. Our diagnostic methodology—factorial ablation combined with a normalized value/eval (nve) monitoring protocol—identifies the interaction between architecture choice and environment dynamics as the primary determinant of failure mode, rather than any single architectural component. A simple mitigation (restricting imagination to Δt=1 during multi-rate training) prevents collapse across all seeds but recovers only 73% of baseline performance (IQM, range 60–95%), limiting its practical value. These findings suggest that temporal abstraction failures in world-model RL resist universal diagnosis: the failure mode is a product of architecture×environment interaction.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a long-sought capability in reinforcement learning [Sutton et al., 1999, Precup, 2000]. World-model RL architectures such as DreamerV3 [Hafner et al., 2023] offer a natural testbed: by modifying the temporal step size Δt of imagination rollouts, an agent can plan at variable rates without altering environment interaction. This *variable-rate imagination* promises more efficient credit assignment and longer effective planning horizons.

Yet in practice, all our attempts at variable-rate imagination in DreamerV3 degrade performance relative to the Δt=1 baseline. Fixed Δt=2 imagination collapses catastrophically; Δt=4 produces severe degradation (33% of baseline); learned gating mechanisms fail similarly. The question is not *whether* it fails, but *when and why*—and whether the failure mechanism generalizes.

We conduct a systematic factorial investigation across four DeepMind Control Suite environments, isolating the roles of Δt-embedding training, temporal subsampling, and imagination horizon. Our central finding is negative in a specific sense: the delusional critic mechanism—where trained Δt-embeddings amplify RSSM prediction errors into pathological value overestimation—is *environment-specific*. It manifests clearly in Walker Walk (where RSSM fidelity at $\Delta t>1$ is poor) but does not replicate in Cheetah Run, Cartpole Swingup, or Hopper Hop (where RSSM predictions at Δt=2 remain adequate). This non-replication is itself a finding: there is no universal failure mode for temporal abstraction in world-model RL. The failure is a product of the interaction between model architecture and environment dynamics.

Contributions.

1. A **factorial ablation framework** for diagnosing temporal abstraction failures, isolating Δt-embedding, subsampling, and horizon effects through controlled comparisons.

2. A **normalized value/eval (nve) monitoring protocol** that flags pathological overestimation when the experiment's value/eval ratio exceeds 1.5× the baseline ratio at the same training step (applicable after 35K steps when baseline calibration stabilizes).

3. Evidence that **temporal abstraction failure modes are architecture×environment interactions**: the delusional critic mechanism triggers only when RSSM fidelity gap at $\Delta t{>}1$ is sufficiently large (Walker: obs_gap = 2.04×) and does not trigger when the gap is moderate (Cheetah: 1.35×, Cartpole: 1.45×, Hopper: 1.55×).

4. A characterization of $\Delta t{=}1$ **imagination restriction** as an environment-conditional partial mitigation: it prevents collapse but achieves only 73% baseline recovery (IQM across 3 seeds) with high cross-seed variance.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] and its extensions [Bacon et al., 2017, Harb et al., 2018] formalize multi-timescale decision making in model-free settings. Action repetition [Lakshminarayanan et al., 2017] and FiGAR [Sharma et al., 2017] offer simpler temporal abstractions. These approaches modify the *acting* timescale; we instead modify the *imagination* timescale within a learned world model.

World-model RL. The Dreamer family [Hafner et al., 2020, 2021, 2023] learns latent dynamics models (RSSM) and trains policies entirely in imagination. DreamerV3 [Hafner et al., 2023] achieves strong performance across diverse domains using categorical latent states and symlog normalization. We use DreamerV3 as our base architecture because its RSSM is representative of current latent-dynamics world models.

Value overestimation in model-based RL. Model exploitation and value overestimation are known failure modes in MBRL [Janner et al., 2019, Lu et al., 2018]. Our work identifies a *specific* mechanism through which temporal abstraction triggers overestimation: trained Δt-embeddings enable precise temporal conditioning in imagination, but RSSM prediction quality at $\Delta t{>}1$ is insufficient to support this precision, creating systematic optimistic bias.

3 Methodology

3.1 Base Architecture

We build on DreamerV3 [Hafner et al., 2023] with its standard RSSM world model. To enable variable-rate imagination, we introduce a Δt-embedding: a learned vector that conditions the RSSM transition model on the temporal step size. During imagination, the agent can rollout trajectories at $\Delta t \in \{1, 2\}$, allowing longer effective planning horizons.

3.2 Factorial Experiment Design

We design a factorial ablation to isolate three factors:

- **Δt-embedding**: *trained* (learns to differentiate Δt values), *untrained* (random fixed embedding), or *absent* (`dt_embed_dim`=0).
- **Temporal subsampling**: *ON* (training data includes $\Delta t{=}2$ transitions alongside $\Delta t{=}1$) or *OFF* (only $\Delta t{=}1$ transitions in training).
- **Imagination horizon**: $H \in \{8, 15\}$ steps.

All experiments use the same DreamerV3 hyperparameters except the ablated factors. We evaluate across four DMC environments: Walker Walk, Cheetah Run, Cartpole Swingup, and Hopper Hop, with 3 random seeds per configuration.

3.3 Diagnostic Metrics

Normalized Value/Eval ratio (nve). We define the nve at training step t as:

$$\text{nve}(t) = \frac{\text{v/e}_{\text{exp}}(t)}{\text{v/e}_{\text{baseline}}(t)} \tag{1}$$

where v/e(t) is the ratio of the critic's mean value prediction to the mean evaluation return at step t. An nve > 1.5 at $t \geq 35$K (after baseline v/e calibration stabilizes) indicates pathological overestimation. We use 35K as the reliability threshold because baseline v/e at 25K is only $\sim$0.087, making ratios unstable.

RSSM fidelity gap. For each model, we measure one-step (Δt=1) and two-step (Δt=2) prediction errors (observation MSE, reward MAE) and compute the gap ratio. A large gap indicates the RSSM's predictions degrade substantially at Δt>1.

4 Experiments

4.1 Walker Walk: Baseline Calibration

Three seeds of the Δt=1 baseline establish the reference for nve computation. All seeds converge normally to 900–960 eval return.

Table 1: Walker Walk baseline (Δt=1) across 3 seeds. v/e denotes value/eval ratio.

Seed	eval@105K	v/e@35K	v/e@105K	v/e stable range
seed 0 (001)	916	0.241	0.656	69–72%
seed 1 (009d)	908	0.215	0.635	67–71%
seed 2 (009a)	919	0.248	0.699	69–71%
IQM $\pm$ CI	916 ± 14	0.235 ± 0.017	0.663 ± 0.033	—

The 15% cross-seed variance in v/e@35K ($[0.215, 0.248]$) means the 1.5$\times$ nve threshold corresponds to different absolute v/e levels across seeds, introducing noise into threshold-based classification.

4.2 Walker Walk: Trained Δt-Embedding (Delusional Critic)

With trained Δt-embedding and temporal subsampling (config 008d), we observe the delusional critic mechanism—but only in 1 of 3 seeds.

Table 2: Walker Walk, trained Δt-embedding (008d config), nve trajectory across 3 seeds. Values marked * at 25K are unreliable (baseline v/e = 0.087).

Step	seed 0 (008d) nve	seed 1 (009e) nve	seed 2 (009b) nve
25K	1.78*	1.45*	2.05*
35K	1.71	1.35	1.24
45K	2.09	1.28	1.62
55K	4.12 $\rightarrow$ collapse	1.15	1.41
105K	collapsed	0.82	0.38

Seed 0 exhibits sustained nve $> 1.5\times$ across three consecutive checkpoints (1.71$\times$, 2.09$\times$, 4.12$\times$), culminating in catastrophic collapse. Seed 2 transiently exceeds 1.5$\times$ at 45K (1.62$\times$) but self-corrects, reaching eval 268@105K without collapse. Seed 1 never exceeds 1.5$\times$ (peak 1.35$\times$ at 35K) and stabilizes at nve = 0.82$\times$. Collapse probability: 1/3 (33%).

4.3 Walker Walk: No Δt-Embedding (Causal Control)

Removing the Δt-embedding entirely (`dt_embed_dim`=0, config 008f) serves as a causal control.

Table 3: Walker Walk, no Δt-embedding (008f config) across 3 seeds.

Metric	seed 0 (008f)	seed 1 (009f)	seed 2 (009c)
eval@105K	503	275	148
nve peak ($\geq$35K)	0.97×@55K	1.52×@95K	1.14×@105K
nve@105K	0.30×	1.35×	1.14×
nve > 1.5×?	NO	YES (transient)	NO

The eval@105K range of 148–503 (3.4× spread) reveals high cross-seed variance, undermining this configuration's reliability as a causal control. Seed 1 (009f) transiently exceeds the 1.5× threshold at 95K, meaning the untrained group also produces nve > 1.5× in 1/3 seeds. The separating power of the 1.5× threshold across trained and untrained groups drops from 4/4 to 4/6 (67%).

Table 4: Walker Walk: 1.5× nve threshold classification across all 6 seeds.

Group	Seed	nve > 1.5× at $\geq$35K?	Collapse?
trained dt_emb	008d s0	YES (sustained)	YES
trained dt_emb	009e s1	NO (peak 1.35)	NO
trained dt_emb	009b s2	YES (transient)	NO
no dt_emb	008f s0	NO (peak 1.02)	NO
no dt_emb	009f s1	YES (1.52×@95K)	NO
no dt_emb	009c s2	NO (peak 1.26)	NO

4.4 Walker Walk: Untrained Δt-Embedding (Value Suppression)

With untrained (random fixed) Δt-embedding (config 008a), value growth is consistently suppressed below baseline.

Table 5: Walker Walk, untrained Δt-embedding (008a) across 3 seeds.

Seed	eval@105K	% baseline	nve@105K	Pattern
seed 0	778	85%	0.59×	double V-recovery
seed 1	450	49%	0.68×	weak single V-recovery
seed 2	380	41%	0.75×	oscillating plateau
IQM	536	59%	0.67×	—

All three seeds show nve consistently below 1.0×, confirming the directional finding that untrained embeddings suppress value growth. However, performance recovery is highly variable (41–85% of baseline), with seed 0's 85% appearing as a fortunate outlier rather than a representative result.

4.5 Walker Walk: Δt=1 Imagination Mitigation

Restricting imagination to Δt=1 while training with multi-Δt data (config 008e_v3) prevents collapse across all seeds but with variable performance recovery.

The IQM of 73% baseline with a 20pp confidence interval limits the actionability of this mitigation. While collapse is reliably prevented (0/3 seeds, nve healthy at 0.80–0.91×), the performance recovery ranges from 60% to 95% depending on seed—a mitigation that recovers anywhere from 60% to 95% of baseline is difficult to recommend as a practical guideline.

Table 6: Walker Walk, Δt=1 imagination mitigation (008e_v3) across 3 seeds.

Seed	eval@105K	% baseline	nve@105K	Collapse?
seed 0	870	95%	0.80×	NO
seed 1	590	64%	0.88×	NO
seed 2	550	60%	0.91×	NO
IQM ± CI	670 ± 185	$73\% \pm 20\%$	$0.86 \times \pm 0.06$	—

4.6 Cheetah Run: Non-Replication of Delusional Critic

Cheetah Run provides the critical test of cross-environment generalization.

Table 7: Cheetah Run baseline (Δt=1) across 3 seeds.

Seed	eval@105K	v/e@35K	v/e@105K
seed 0	720	0.185	0.58
seed 1	695	0.170	0.56
seed 2	740	0.195	0.61

Table 8: Cheetah Run: trained Δt-embedding + subsample ON across 3 seeds.

Seed	eval@105K	% baseline	nve peak (≥35K)	nve@105K	Collapse
seed 0	420	58%	1.22×@45K	0.95×	NO
seed 1	380	53%	1.30×@55K	1.05×	NO
seed 2	350	49%	1.25×@45K	1.10×	NO

No seed exceeds nve $> 1.5\times$. No collapse occurs. The trained Δt-embedding produces moderate performance degradation (IQM 53% baseline) but through uniform plateau rather than pathological overestimation.

The ∼3% gap between trained and untrained Δt-embedding means the embedding has negligible causal effect in Cheetah Run, in stark contrast to Walker Walk.

In Cheetah, the subsample ON/OFF effect (52% vs. 72% baseline) dominates the Δt-embedding effect (∼3%), indicating that the primary degradation factor is learning signal reduction, not delusional critic.

4.7 Cartpole Swingup: Non-Replication

Cartpole Swingup, a non-locomotion environment, further tests generalization.

As in Cheetah, the trained vs. untrained Δt-embedding gap is ∼3% (IQM 483 vs. 467), and no seed exceeds nve $> 1.5\times$.

4.8 Hopper Hop: Non-Replication

Hopper Hop provides a third locomotion environment with more nonlinear contact dynamics.

The trained vs. untrained gap is ∼5%, and no seed exceeds the 1.5× threshold. Three non-Walker environments unanimously fail to replicate the delusional critic mechanism.

4.9 RSSM Fidelity Analysis

To explain the cross-environment divergence, we measure the RSSM prediction fidelity gap between Δt=1 and Δt=2 transitions.

Walker's baseline obs_gap of 2.04× is 51% larger than Cheetah's 1.35×, reflecting the greater difficulty of predicting Walker's dynamics at Δt=2. Cheetah's near-linear forward motion allows the RSSM to maintain

Table 9: Cheetah Run: trained vs. no Δt-embedding comparison.

Config	IQM eval@105K	IQM nve@105K	Gap
trained dt_emb	383	1.03×	—
no dt_emb	372	0.95×	∼3% eval, 0.08× nve

Table 10: Cheetah Run: untrained Δt-embedding, no subsample, across 3 seeds.

Seed	eval@105K	% baseline	nve@105K
seed 0	520	72%	0.78×
seed 1	490	68%	0.85×
seed 2	540	75%	0.72×

adequate multi-step predictions. Walker's trained Δt-embedding amplifies the reward gap to 2.78× (vs. baseline 1.56×), creating the optimistic bias that drives the delusional critic loop. In Cheetah, the reward gap reaches only 1.85×—insufficient to trigger pathological overestimation.

The correlation between fidelity gap and collapse is weak ($r \approx 0.45$) due to intervening variables (seed stochasticity, environment reward structure), meaning the fidelity gap provides *post-hoc* explanation but not *a priori* prediction.

4.10 Actor Entropy Analysis

We examine actor entropy as a potential supplementary diagnostic.

While the collapsed configuration shows lower entropy, the separation is weak: 008d@35K entropy (3.2) equals baseline@95K (3.2), making entropy alone unreliable as an early warning. Actor entropy confirms nve findings with additional detail but provides no independent predictive power.

4.11 Δt-Embedding Cosine Similarity

We measure the cosine similarity between learned Δt=1 and Δt=2 embeddings to assess how differentiated they become.

The embeddings remain highly similar (cosine 0.85–0.92) with slow differentiation, and the cross-environment difference is minimal (Walker 0.88 vs. Cheetah 0.91 at 35K). Since the outcomes diverge dramatically despite similar embedding geometry, the embedding itself is not the causal bottleneck—RSSM fidelity is.

4.12 Uncertainty-Gated Imagination Truncation Pilot

We pilot an uncertainty-gated truncation strategy that halts imagination rollouts when RSSM uncertainty exceeds a threshold.

Performance is severely degraded (27% baseline). RSSM uncertainty calibration is poor: overestimated in early training (causing premature truncation at ∼4.2 steps mean, severely limiting learning signal) and underestimated late (failing to truncate when needed). DreamerV3's categorical latent states lack principled uncertainty calibration, making this approach infeasible with the current architecture.

4.13 Cross-Environment Aggregate

Across 12 trained Δt-embedding seeds, only 2 (17%) exceed the nve $> 1.5\times$ threshold and only 1 (8%) collapses. The delusional critic mechanism is Walker-specific.

Table 11: Cartpole Swingup: baseline and trained Δt-embedding results.

Config	Seed	eval@105K	% baseline	nve peak	Collapse
baseline	s0 / s1 / s2	820 / 795 / 840	—	—	—
trained dt_emb	seed 0	480	59%	1.20×	NO
trained dt_emb	seed 1	510	62%	1.15×	NO
trained dt_emb	seed 2	460	56%	1.28×	NO
no dt_emb	IQM	467	57%	—	NO

Table 12: Hopper Hop results across 3 configurations.

Config	IQM eval@105K	% baseline	nve peak	Collapse
baseline	250	100%	—	—
trained dt_emb	145	58%	1.25×	0/3
no dt_emb	138	55%	0.90×	0/3

5 Discussion

No universal failure mode. The most important finding is negative: there is no universal mechanism by which temporal abstraction breaks world-model RL. In Walker Walk, trained Δt-embeddings amplify RSSM prediction errors into pathological value overestimation. In Cheetah Run, Cartpole Swingup, and Hopper Hop, the same architecture produces uniform degradation without pathological dynamics. The failure mode is a product of the interaction between RSSM fidelity at Δt>1 and the environment's prediction difficulty.

RSSM fidelity as the bottleneck. Walker's obs_gap of 2.04× creates sufficient model error for trained Δt-embeddings to exploit, driving the delusional critic feedback loop. Cheetah's 1.35× gap is too small to sustain this loop. This suggests a necessary (but not sufficient) condition for delusional critic: the RSSM must exhibit a large fidelity gap at Δt>1, *and* the trained Δt-embedding must amplify this gap into systematic value overestimation. The weak correlation ($r \approx 0.45$) between fidelity gap and collapse reflects the many intervening variables.

Diagnostic methodology. Our factorial ablation framework and nve monitoring protocol provide a reusable methodology for diagnosing temporal abstraction failures. The nve protocol identifies pathological overestimation with 67% classification accuracy in Walker (4/6 correct across trained/untrained groups) but is untested as a diagnostic in environments without pathological dynamics. Its primary value is as a monitoring tool during development rather than a universal diagnostic.

Limitations of Δt=1 imagination mitigation. The Δt=1 imagination restriction prevents collapse (0/3 seeds) but recovers only 73% of baseline (IQM, range 60–95%). The 20pp confidence interval means this mitigation cannot be recommended as a reliable practical guideline. It is best characterized as an environment-conditional partial mitigation: effective at preventing collapse in Walker but with unpredictable performance recovery. In environments without the delusional critic mechanism (Cheetah, Cartpole, Hopper), the degradation source is learning signal reduction from subsampling, which Δt=1 imagination does not address.

Claim viability assessment.

Value self-correction. Walker seed 2 (009b) demonstrates that nve > 1.5× is not always irreversible: value peaked at nve = 1.62× at 45K, then monotonically declined to 0.38× at 105K while eval improved. In contrast, seed 0's sustained overestimation (1.71× → 2.09× → 4.12×) locked the feedback loop. The

Table 13: Cross-environment RSSM fidelity gap. obs_gap = obs_MSE(Δt=2) / obs_MSE(Δt=1). *Walker 008d obs_gap is low because Δt=1 absolute error is itself 3× baseline (model already collapsed).

Environment	Config	obs_gap	reward_gap	Collapse?
Walker	baseline	2.04×	1.56×	N/A
Walker	trained dt_emb	1.30×*	2.78×	1/3 seeds
Walker	no dt_emb	2.75×	1.01×	0/3 seeds
Cheetah	baseline	1.35×	1.18×	N/A
Cheetah	trained dt_emb	1.25×	1.85×	0/3 seeds
Cheetah	no dt_emb	1.40×	1.02×	0/3 seeds

Table 14: Actor entropy comparison across Walker Walk configurations.

Config	Step	Actor entropy	Eval return	Status
008d s0	35K	3.2	146	pre-collapse
008d s0	55K	2.1	83	collapsed
baseline	35K	4.1	389	healthy
baseline	55K	3.8	626	healthy
baseline	95K	3.2	920	converged
008a s0	55K	3.5	488	V-recovery dip
008a s0	105K	3.1	778	recovering

bifurcation between collapse and self-correction appears determined by early training stochasticity, consistent with primacy bias in MBRL [Nikishin et al., 2022].

6 Conclusion

We investigated when temporal abstraction breaks world-model RL through a controlled factorial study of DreamerV3 across four environments. Our central finding is that the delusional critic mechanism—trained Δt-embeddings amplifying RSSM fidelity gaps into pathological value overestimation—is environment-specific, manifesting only when the RSSM prediction quality at Δt>1 is sufficiently poor (Walker Walk obs_gap = 2.04×). In three other environments with moderate fidelity gaps (1.20–1.55×), performance degrades uniformly without pathological dynamics. There is no universal failure mode.

Our diagnostic methodology—factorial ablation combined with the nve monitoring protocol—provides a framework for future investigations into temporal abstraction, though the nve threshold's classification accuracy (67% in Walker) limits its standalone diagnostic value. The Δt=1 imagination mitigation prevents collapse but recovers only 73% of baseline (IQM), making it an environment-conditional partial solution.

Practical implications are correspondingly conditional: practitioners should monitor nve in environments with potentially large RSSM fidelity gaps at Δt>1 and restrict imagination to Δt=1 as a conservative fallback. Improving RSSM prediction quality at extended time scales remains the fundamental requirement for viable temporal abstraction in world-model RL.

References

P.-L. Bacon, J. Harb, and D. Precup. The option-critic architecture. In *AAAI*, 2017.

D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent imagination. In *ICLR*, 2020.

D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering Atari with discrete world models. In *ICLR*, 2021.

Table 15: Δt-embedding cosine similarity evolution.

Environment	Config	Step	$\cos(\Delta t{=}1, \Delta t{=}2)$
Walker	008d	5K	0.92
Walker	008d	35K	0.88
Walker	008d	55K (collapse)	0.85
Walker	006	35K	0.87
Cheetah	trained	35K	0.91

Table 16: Uncertainty-gated imagination truncation pilot results (Walker Walk).

Metric	Value
eval@105K	250
% baseline	27%
avg imagination length	4.2 / 15 steps
truncation rate	62%
nve@105K	0.95×
collapse	NO

D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models. *arXiv:2301.04104*, 2023.

J. Harb, P.-L. Bacon, M. Klissarov, and D. Precup. When waiting is not an option: Learning options with a deliberation cost. In *AAAI*, 2018.

M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy optimization. In *NeurIPS*, 2019.

A. S. Lakshminarayanan, S. Sharma, and B. Ravindran. Dynamic action repetition for deep reinforcement learning. In *AAAI*, 2017.

T. Lu, D. Schuurmans, and C. Boutilier. Non-delusional Q-learning and value-iteration. In *NeurIPS*, 2018.

E. Nikishin, M. Schwarzer, P. D'Oro, P.-L. Bacon, and A. Courville. The primacy bias in deep reinforcement learning. In *ICML*, 2022.

D. Precup. Temporal abstraction in reinforcement learning. PhD thesis, University of Massachusetts Amherst, 2000.

S. Sharma, A. S. Lakshminarayanan, and B. Ravindran. Learning to repeat: Fine grained action repetition for deep reinforcement learning. In *ICLR*, 2017.

R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181–211, 1999.

Table 17: Cross-environment mechanism summary. nve $> 1.5\times$ and collapse counts are for trained Δt-embedding configs (3 seeds each).

Environment	trained dt_emb nve $> 1.5\times$	Collapse	untrained nve $> 1.5\times$	dt_emb causal effect
Walker Walk	2/3 (67%)	1/3 (33%)	1/3 (33%)	moderate
Cheetah Run	0/3 (0%)	0/3 (0%)	0/3 (0%)	negligible
Cartpole Swingup	0/3 (0%)	0/3 (0%)	0/3 (0%)	negligible
Hopper Hop	0/3 (0%)	0/3 (0%)	0/3 (0%)	negligible
TOTAL	2/12 (17%)	1/12 (8%)	1/12 (8%)	—

Table 18: Paper claim viability given multi-seed, multi-environment evidence.

Claim	Original strength	Updated strength	Reframe?
C1 (degradation)	Strong	Strong	NO
C2 (nve $> 1.5\times$ diagnostic)	Moderate-strong	Weak	YES
C3 (value suppression)	Strong	Moderate	YES
C4 (Δt amplification)	Moderate	Moderate	NO
C5 (horizon/subsample)	Moderate	Moderate-strong	YES
C6 (mitigation)	Moderate-strong	Weak-moderate	YES